

times the class dual to the 4-cell. We claim that $b_{jk} = a_{jk}$ for $j < k$ and $b_{jj} = a_j$. If φ is one of the generators η_i or $[i_j, i_k]$ this is clear, since if $\varphi = \eta_j$ then X_φ is the wedge sum of $\mathbb{CP}^2$ with $m - 1$ 2-spheres, while if $\varphi = [i_j, i_k]$ then X_φ is the wedge sum of $S_j^2 \times S_k^2$ with $m - 2$ 2-spheres. The claim is also true when φ is $-\eta_j$ or $-[i_j, i_k]$ since changing the sign of φ amounts to composing φ with a reflection of S^3 , and this changes the generator of $H^4(X_\varphi; \mathbb{Z})$ to its negative. The general case now follows by induction from the assertion that the matrix (b_{jk}) for $X_{\varphi+\psi}$ is the sum of the corresponding matrices for X_φ and X_ψ . This assertion can be proved as follows. By attaching two 4-cells to $\bigvee_j S_j^2$ by φ and ψ we obtain a complex $X_{\varphi,\psi}$ which we can view as $X_\varphi \cup X_\psi$. There is a quotient map $q: X_{\varphi+\psi} \rightarrow X_{\varphi,\psi}$ that is a homeomorphism on the 2-skeleton and collapses the closure of an equatorial 3-disk in the 4-cell of $X_{\varphi+\psi}$ to a point. The induced map $q^*: H^4(X_{\varphi,\psi}) \rightarrow H^4(X_{\varphi+\psi})$ sends each of the two generators corresponding to the 4-cells of $X_{\varphi,\psi}$ to a generator, and the assertion follows.

Now suppose X_φ and X_ψ have isomorphic cup product rings. This means bases for $H^*(X_\varphi; \mathbb{Z})$ and $H^*(X_\psi; \mathbb{Z})$ can be chosen so that the matrices specifying the cup product $H^2 \times H^2 \rightarrow H^4$ with respect to these bases are the same. The preceding proposition says that any choice of basis can be realized as the dual basis to a cell structure on a CW complex homotopy equivalent to the given complex. Therefore we may assume the matrices (b_{jk}) for X_φ and X_ψ are the same. By what we have shown in the preceding paragraph, this means φ and ψ are homotopic, hence X_φ and X_ψ are homotopy equivalent.

For the statement about simply-connected closed 4-manifolds, Corollaries A.8 and A.9 and Proposition A.11 in the Appendix say that such a manifold M has the homotopy type of a CW complex with finitely generated homology groups. Then Poincaré duality and the universal coefficient theorem imply that the only nontrivial homology groups $H_i(M)$ are $\mathbb{Z}$ for $i = 0, 4$ and $\mathbb{Z}^m$ for $i = 2$, for some $m \geq 0$. $\square$

This result and the example preceding it are special cases of a homotopy classification by Whitehead of simply-connected CW complexes with positive-dimensional cells in three adjacent dimensions n , $n + 1$, and $n + 2$; see [Baues 1996] for a full treatment of this.

4.D Cohomology of Fiber Bundles

While the homotopy groups of the three spaces in a fiber bundle fit into a long exact sequence, the relation between their homology or cohomology groups is much more complicated. The Künneth formula shows that there are some subtleties even for a product bundle, and for general bundles the machinery of spectral sequences,

developed in [SSAT], is required. In this section we will describe a few special sorts of fiber bundles where more elementary techniques suffice. As applications we calculate the cohomology rings of some important spaces closely related to Lie groups. In particular we find a number of spaces with exterior and polynomial cohomology rings.

The Leray-Hirsch Theorem

This theorem will be the basis for all the other results in this section. It gives hypotheses sufficient to guarantee that a fiber bundle has cohomology very much like that of a product bundle.

Theorem 4D.1. *Let $F \rightarrow E \xrightarrow{p} B$ be a fiber bundle such that, for some commutative coefficient ring R :*

- (a) $H^n(F; R)$ is a finitely generated free R -module for each n .
- (b) There exist classes $c_j \in H^{k_j}(E; R)$ whose restrictions $i^*(c_j)$ form a basis for $H^*(F; R)$ in each fiber F , where $i: F \rightarrow E$ is the inclusion.

Then the map $\Phi: H^(B; R) \otimes_R H^*(F; R) \rightarrow H^*(E; R)$, $\sum_{ij} b_i \otimes i^*(c_j) \mapsto \sum_{ij} p^*(b_i) \cup c_j$, is an isomorphism.*

In other words, $H^*(E; R)$ is a free $H^*(B; R)$ -module with basis $\{c_j\}$, where we view $H^*(E; R)$ as a module over the ring $H^*(B; R)$ by defining scalar multiplication by $bc = p^*(b) \cup c$ for $b \in H^*(B; R)$ and $c \in H^*(E; R)$.

In the case of a product $E = B \times F$ with $H^*(F; R)$ free over R , we can pull back a basis for $H^*(F; R)$ via the projection $E \rightarrow F$ to obtain the classes c_j . Thus the Leray-Hirsch theorem generalizes the version of the Künneth formula involving cup products, Theorem 3.16, at least as far as the additive structure and the module structure over $H^*(B; R)$ are concerned. However, the Leray-Hirsch theorem does not assert that the isomorphism $H^*(E; R) \approx H^*(B; R) \otimes_R H^*(F; R)$ is a ring isomorphism, and in fact this need not be true, as we shall see by an example later.

An example of a bundle where the classes c_j do not exist is the Hopf bundle $S^1 \rightarrow S^3 \rightarrow S^2$, since $H^*(S^3) \not\approx H^*(S^2) \otimes H^*(S^1)$.

Proof: We first prove the result for finite-dimensional CW complexes B by induction on their dimension. The case that B is 0-dimensional is trivial. For the induction step, suppose B has dimension n , and let $B' \subset B$ be the subspace obtained by deleting a point x_α from the interior of each n -cell e_α^n of B . Let $E' = p^{-1}(B')$. Then we have a commutative diagram, with coefficients in R understood:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & H^*(B, B') \otimes_R H^*(F) & \longrightarrow & H^*(B) \otimes_R H^*(F) & \longrightarrow & H^*(B') \otimes_R H^*(F) \longrightarrow \cdots \\ & & \downarrow \Phi & & \downarrow \Phi & & \downarrow \Phi \\ \cdots & \longrightarrow & H^*(E, E') & \longrightarrow & H^*(E) & \longrightarrow & H^*(E') \longrightarrow \cdots \end{array}$$

The map Φ on the left is defined exactly as in the absolute case, using the relative cup product $H^*(E, E') \otimes_R H^*(E) \rightarrow H^*(E, E')$. The first row of the diagram is exact since

tensoring with a free module preserves exactness. The second row is of course exact also. The commutativity of the diagram follows from the evident naturality of Φ in the case of the two squares shown. For the other square involving coboundary maps, if we start with an element $b \otimes i^*(c_j) \in H^*(B') \otimes_R H^*(F)$ and map this horizontally we get $\delta b \otimes i^*(c_j)$ which maps vertically to $p^*(\delta b) \smile c_j$, whereas if we first map vertically we get $p^*(b) \smile c_j$ which maps horizontally to $\delta(p^*(b) \smile c_j) = \delta p^*(b) \smile c_j = p^*(\delta b) \smile c_j$ since $\delta c_j = 0$.

The space B' deformation retracts onto the skeleton B^{n-1} , and the following lemma implies that the inclusion $p^{-1}(B^{n-1}) \hookrightarrow E'$ is a weak homotopy equivalence, hence induces an isomorphism on all cohomology groups:

Lemma 4D.2. *Given a fiber bundle $p: E \rightarrow B$ and a subspace $A \subset B$ such that (B, A) is k -connected, then $(E, p^{-1}(A))$ is also k -connected.*

Proof: For a map $g: (D^i, \partial D^i) \rightarrow (E, p^{-1}(A))$ with $i \leq k$, there is by hypothesis a homotopy $f_t: (D^i, \partial D^i) \rightarrow (B, A)$ of $f_0 = pg$ to a map f_1 with image in A . The homotopy lifting property then gives a homotopy $g_t: (D^i, \partial D^i) \rightarrow (E, p^{-1}(A))$ of g to a map with image in $p^{-1}(A)$. $\square$

The theorem for finite-dimensional B will now follow by induction on n and the five-lemma once we show that the left-hand Φ in the diagram is an isomorphism.

By the fiber bundle property there are open disk neighborhoods $U_\alpha \subset e_\alpha^n$ of the points x_α such that the bundle is a product over each U_α . Let $U = \bigcup_\alpha U_\alpha$ and let $U' = U \cap B'$. By excision we have $H^*(B, B') \approx H^*(U, U')$, and $H^*(E, E') \approx H^*(p^{-1}(U), p^{-1}(U'))$. This gives a reduction to the problem of showing that the map $\Phi: H^*(U, U') \otimes_R H^*(F) \rightarrow H^*(U \times F, U' \times F)$ is an isomorphism. For this we can either appeal to the relative Künneth formula in Theorem 3.21 or we can argue again by induction, applying the five-lemma to the diagram with (B, B') replaced by (U, U') , induction implying that the theorem holds for U and U' since they deformation retract onto complexes of dimensions 0 and $n-1$, respectively, and by the lemma we can restrict to the bundles over these complexes.

Next there is the case that B is an infinite-dimensional CW complex. Since (B, B^n) is n -connected, the lemma implies that the same is true of $(E, p^{-1}(B^n))$. Hence in the commutative diagram at the right the horizontal maps are isomorphisms below dimension n . Then the fact that the right-hand Φ is an isomorphism, as we have already shown, implies that the left-hand Φ is an isomorphism below dimension n . Since n is arbitrary, this gives the theorem for all CW complexes B .

$$\begin{array}{ccc} H^*(B) \otimes_R H^*(F) & \longrightarrow & H^*(B^n) \otimes_R H^*(F) \\ \downarrow \Phi & & \downarrow \Phi \\ H^*(E) & \longrightarrow & H^*(p^{-1}(B^n)) \end{array}$$

To extend to the case of arbitrary base spaces B we need the notion of a **pull-back bundle** which is used quite frequently in bundle theory. Given a fiber bundle

$p: E \rightarrow B$ and a map $f: A \rightarrow B$, let $f^*(E) = \{(a, e) \in A \times E \mid f(a) = p(e)\}$, so there is a commutative diagram as at the right, where the two maps from $f^*(E)$ are $(a, e) \mapsto a$ and $(a, e) \mapsto e$. It is a simple exercise to verify that the projection $f^*(E) \rightarrow A$ is a fiber bundle with the same fiber as $E \rightarrow B$, since a local trivialization of $E \rightarrow B$ over $U \subset B$ gives rise to a local trivialization of $f^*(E) \rightarrow A$ over $f^{-1}(U)$.

$$\begin{array}{ccc} f^*(E) & \longrightarrow & E \\ \downarrow & & \downarrow p \\ A & \xrightarrow{f} & B \end{array}$$

If $f: A \rightarrow B$ is a CW approximation to an arbitrary base space B , then $f^*(E) \rightarrow E$ induces an isomorphism on homotopy groups by the five-lemma applied to the long exact sequences of homotopy groups for the two bundles $E \rightarrow B$ and $f^*(E) \rightarrow A$ with fiber F . Hence $f^*(E) \rightarrow E$ is also an isomorphism on cohomology. The classes c_j pull back to classes in $H^*(f^*(E); R)$ which still restrict to a basis in each fiber, and so the naturality of Φ reduces the theorem for $E \rightarrow B$ to the case of $f^*(E) \rightarrow A$. $\square$

Corollary 4D.3. (a) $H^*(U(n); \mathbb{Z}) \approx \Lambda_{\mathbb{Z}}[x_1, x_3, \dots, x_{2n-1}]$, the exterior algebra on generators x_i of odd dimension i .

(b) $H^*(SU(n); \mathbb{Z}) \approx \Lambda_{\mathbb{Z}}[x_3, x_5, \dots, x_{2n-1}]$.

(c) $H^*(Sp(n); \mathbb{Z}) \approx \Lambda_{\mathbb{Z}}[x_3, x_7, \dots, x_{4n-1}]$.

This rather simple structure is in marked contrast with the cohomology of $O(n)$ and $SO(n)$ which is considerably more complicated, as shown in §3.D.

Proof: For (a), assume inductively that the result holds for $U(n-1)$. By considering the bundle $U(n-1) \rightarrow U(n) \rightarrow S^{2n-1}$ we see that the pair $(U(n), U(n-1))$ is $(2n-2)$ -connected, so $H^i(U(n); \mathbb{Z}) \rightarrow H^i(U(n-1); \mathbb{Z})$ is onto for $i \leq 2n-3$ and the classes $x_1, \dots, x_{2n-3} \in H^*(U(n-1); \mathbb{Z})$ given by induction are the restrictions of classes $c_1, \dots, c_{2n-3} \in H^*(U(n); \mathbb{Z})$. The products of distinct x_i 's form a basis for $H^*(U(n-1); \mathbb{Z}) \approx \Lambda_{\mathbb{Z}}[x_1, \dots, x_{2n-3}]$, and these products are restrictions of the corresponding products of c_i 's, so the Leray-Hirsch theorem applies, yielding $H^*(U(n); \mathbb{Z}) \approx H^*(U(n-1); \mathbb{Z}) \otimes H^*(S^{2n-1}; \mathbb{Z})$. In view of the commutativity property of cup product, this tensor product is the exterior algebra on odd-dimensional generators $x_1, \dots, x_{2n-1}$.

The same proof works for $Sp(n)$ using the bundle $Sp(n-1) \rightarrow Sp(n) \rightarrow S^{4n-1}$. In the case of $SU(n)$ one uses the bundle $SU(n-1) \rightarrow SU(n) \rightarrow S^{2n-1}$. Since $SU(1)$ is the trivial group, the bundle $SU(1) \rightarrow SU(2) \rightarrow S^3$ shows that $SU(2) = S^3$, so the first generator is x_3 . $\square$

It is illuminating to look more closely at how the homology and cohomology of $O(n)$, $U(n)$, and $Sp(n)$ are related to their bundle structures. For $U(n)$ one has the sequence of bundles

$$\begin{array}{ccccccc} S^1 = U(1) & \hookrightarrow & U(2) & \hookrightarrow & U(3) & \hookrightarrow & \dots \hookrightarrow U(n-1) \hookrightarrow U(n) \\ & & \downarrow & & \downarrow & & \downarrow \\ & & S^3 & & S^5 & & \dots \quad S^{2n-3} \quad S^{2n-1} \end{array}$$

If all these were product bundles, $U(n)$ would be homeomorphic to the product $S^1 \times S^3 \times \cdots \times S^{2n-1}$. In actuality the bundles are nontrivial, but the homology and cohomology of $U(n)$ are the same as for this product of spheres, including the cup product structure. For $Sp(n)$ the situation is quite similar, with the corresponding product of spheres $S^3 \times S^7 \times \cdots \times S^{4n-1}$. For $O(n)$ the corresponding sequence of bundles is

$$\begin{array}{ccccccc} S^0 = O(1) & \hookrightarrow & O(2) & \hookrightarrow & O(3) & \hookrightarrow & \cdots \hookrightarrow O(n-1) \hookrightarrow O(n) \\ & & \downarrow & & \downarrow & & \downarrow \\ & & S^1 & & S^2 & & \cdots \quad S^{n-2} \quad S^{n-1} \end{array}$$

The calculations in §3.D show that $H_*(O(n); \mathbb{Z}_2) \approx H_*(S^0 \times S^1 \times \cdots \times S^{n-1}; \mathbb{Z}_2)$, but with $\mathbb{Z}$ coefficients this no longer holds. Instead, consider the coarser sequence of bundles

$$\begin{array}{ccccccc} S^0 = O(1) & \hookrightarrow & O(3) & \hookrightarrow & O(5) & \hookrightarrow & \cdots \hookrightarrow O(2k-1) \hookrightarrow O(2k) \\ & & \downarrow & & \downarrow & & \downarrow \\ & & V_2(\mathbb{R}^3) & & V_2(\mathbb{R}^5) & & \cdots \quad V_2(\mathbb{R}^{2k-1}) \quad S^{2k-1} \end{array}$$

where the last bundle $O(2k) \rightarrow S^{2k-1}$ is omitted if $n = 2k - 1$. As we remarked at the end of §3.D in the case of $SO(n)$, $O(n)$ has the same integral homology and cohomology groups as if these bundles were products, but the cup product structure for $O(n)$ with $\mathbb{Z}_2$ coefficients is not the same as in this product.

Cohomology of Grassmannians

Here is an important application of the Leray-Hirsch theorem, generalizing the calculation of the cohomology rings of projective spaces:

Theorem 4D.4. *If $G_n(\mathbb{C}^\infty)$ is the Grassmann manifold of n -dimensional vector subspaces of $\mathbb{C}^\infty$, then $H^*(G_n(\mathbb{C}^\infty); \mathbb{Z})$ is a polynomial ring $\mathbb{Z}[c_1, \dots, c_n]$ on generators c_i of dimension $2i$. Similarly, $H^*(G_n(\mathbb{R}^\infty); \mathbb{Z}_2)$ is a polynomial ring $\mathbb{Z}_2[w_1, \dots, w_n]$ on generators w_i of dimension i , and $H^*(G_n(\mathbb{H}^\infty); \mathbb{Z}) \approx \mathbb{Z}[q_1, \dots, q_n]$ with q_i of dimension $4i$.*

The plan of the proof is to apply the Leray-Hirsch theorem to a fiber bundle $F \rightarrow E \xrightarrow{p} G_n(\mathbb{C}^\infty)$ where E has the same cohomology ring as the product of n copies of $\mathbb{CP}^\infty$, a polynomial ring $\mathbb{Z}[x_1, \dots, x_n]$ with each x_i 2-dimensional. The induced map $p^*: H^*(G_n(\mathbb{C}^\infty); \mathbb{Z}) \rightarrow H^*(E; \mathbb{Z})$ will be injective, and we will show that its image consists of the symmetric polynomials in $\mathbb{Z}[x_1, \dots, x_n]$, the polynomials invariant under permutations of the variables x_i . It is a classical theorem in algebra that the symmetric polynomials themselves form a polynomial ring $\mathbb{Z}[\sigma_1, \dots, \sigma_n]$ where σ_i is a certain symmetric polynomial of degree i , namely the sum of all products of i distinct x_j 's. This gives the result for $G_n(\mathbb{C}^\infty)$, and the same argument will also apply in the real and quaternionic cases.

Proof: Define an n -flag in $\mathbb{C}^k$ to be an ordered n -tuple of orthogonal 1-dimensional vector subspaces of $\mathbb{C}^k$. Equivalently, an n -flag could be defined as a chain of vector subspaces $V_1 \subset \cdots \subset V_n$ of $\mathbb{C}^k$ where V_i has dimension i . Why either of these objects should be called a ‘flag’ is not exactly clear, but that is the traditional name. The set of all n -flags in $\mathbb{C}^k$ forms a subspace $F_n(\mathbb{C}^k)$ of the product of n copies of $\mathbb{CP}^{k-1}$. There is a natural fiber bundle

$$F_n(\mathbb{C}^n) \longrightarrow F_n(\mathbb{C}^k) \xrightarrow{p} G_n(\mathbb{C}^k)$$

where p sends an n -tuple of orthogonal lines to the n -plane it spans. The local triviality property can be verified just as was done for the analogous Stiefel bundle $V_n(\mathbb{C}^n) \rightarrow V_n(\mathbb{C}^k) \rightarrow G_n(\mathbb{C}^k)$ in Example 4.53. The case $k = \infty$ is covered by the same argument, and this case will be the bundle $F \rightarrow E \rightarrow G_n(\mathbb{C}^\infty)$ alluded to in the paragraph preceding the proof.

The first step in the proof is to show that $H^*(F_n(\mathbb{C}^\infty); \mathbb{Z}) \approx \mathbb{Z}[x_1, \dots, x_n]$ where x_i is the pullback of a generator of $H^2(\mathbb{CP}^\infty; \mathbb{Z})$ under the map $F_n(\mathbb{C}^\infty) \rightarrow \mathbb{CP}^\infty$ projecting an n -flag onto its i^{th} line. This can be seen by considering the fiber bundle

$$\mathbb{CP}^\infty \longrightarrow F_n(\mathbb{C}^\infty) \xrightarrow{p} F_{n-1}(\mathbb{C}^\infty)$$

where p projects an n -flag onto the $(n-1)$ -flag obtained by ignoring its last line. The local triviality property can be verified by the argument in Example 4.54. The Leray-Hirsch theorem applies since the powers of x_n restrict to a basis for $H^*(\mathbb{CP}^\infty; \mathbb{Z})$ in the fibers $\mathbb{CP}^\infty$, each fiber being the space of lines in a vector subspace $\mathbb{C}^\infty$ of the standard $\mathbb{C}^\infty$. The elements x_i for $i < n$ are the pullbacks via p of elements of $H^*(F_{n-1}(\mathbb{C}^\infty); \mathbb{Z})$ defined in the same way. By induction $H^*(F_{n-1}(\mathbb{C}^\infty); \mathbb{Z})$ is a polynomial ring on these elements. From the Leray-Hirsch theorem we conclude that the products of powers of the x_i ’s for $1 \leq i \leq n$ form an additive basis for $H^*(F_n(\mathbb{C}^\infty); \mathbb{Z})$, hence this ring is the polynomial ring on the x_i ’s.

There is a corresponding result for $F_n(\mathbb{C}^k)$, that $H^*(F_n(\mathbb{C}^k); \mathbb{Z})$ is free with basis the monomials $x_1^{i_1} \cdots x_n^{i_n}$ with $i_j \leq k-j$ for each j . This is proved in exactly the same way, using induction on n and the fiber bundle $\mathbb{CP}^{k-n} \rightarrow F_n(\mathbb{C}^k) \rightarrow F_{n-1}(\mathbb{C}^k)$. Thus the cohomology groups of $F_n(\mathbb{C}^k)$ are isomorphic to those of $\mathbb{CP}^{k-1} \times \cdots \times \mathbb{CP}^{k-n}$.

After these preliminaries we can start the main argument, using the fiber bundle $F_n(\mathbb{C}^n) \rightarrow F_n(\mathbb{C}^\infty) \xrightarrow{p} G_n(\mathbb{C}^\infty)$. The preceding calculations show that the Leray-Hirsch theorem applies, so $H^*(F_n(\mathbb{C}^\infty); \mathbb{Z})$ is a free module over $H^*(G_n(\mathbb{C}^\infty); \mathbb{Z})$ with basis the monomials $x_1^{i_1} \cdots x_n^{i_n}$ with $i_j \leq n-j$ for each j . In particular, since 1 is among the basis elements, the homomorphism p^* is injective and its image is a direct summand of $H^*(F_n(\mathbb{C}^\infty); \mathbb{Z})$. It remains to show that the image of p^* is exactly the symmetric polynomials.

To show that the image of p^* is contained in the symmetric polynomials, consider a map $\pi: F_n(\mathbb{C}^\infty) \rightarrow F_n(\mathbb{C}^\infty)$ permuting the lines in each n -flag according to a given

permutation of the numbers $1, \dots, n$. The induced map π^* on $H^*(F_n(\mathbb{C}^\infty); \mathbb{Z}) \approx \mathbb{Z}[x_1, \dots, x_n]$ is the corresponding permutation of the variables x_i . Since permuting the lines in an n -flag has no effect on the n -plane they span, we have $p\pi = p$, hence $\pi^*p^* = p^*$, which says that polynomials in the image of p^* are invariant under permutations of the variables.

As remarked earlier, the symmetric polynomials in $\mathbb{Z}[x_1, \dots, x_n]$ form a polynomial ring $\mathbb{Z}[\sigma_1, \dots, \sigma_n]$ where σ_i has degree i . We have shown that the image of p^* is a direct summand, so to show that p^* maps onto the symmetric polynomials it will suffice to show that the graded rings $H^*(G_n(\mathbb{C}^\infty); \mathbb{Z})$ and $\mathbb{Z}[\sigma_1, \dots, \sigma_n]$ have the same rank in each dimension, where the rank of a finitely generated free abelian group is the number of $\mathbb{Z}$ summands.

For a graded free $\mathbb{Z}$ -module $A = \bigoplus_i A_i$, define its **Poincaré series** to be the formal power series $p_A(t) = \sum_i a_i t^i$ where a_i is the rank of A_i , which we assume to be finite for all i . The basic formula we need is that $p_{A \otimes B}(t) = p_A(t) p_B(t)$, which is immediate from the definition of the graded tensor product.

In the case at hand all nonzero cohomology is in even dimensions, so let us simplify notation by taking A_i to be the $2i$ -dimensional cohomology of the space in question. Since the Poincaré series of $\mathbb{Z}[x]$ is $\sum_i t^i = (1-t)^{-1}$, the Poincaré series of $H^*(F_n(\mathbb{C}^\infty); \mathbb{Z})$ is $(1-t)^{-n}$. For $H^*(F_n(\mathbb{C}^n); \mathbb{Z})$ the Poincaré series is

$$(1+t)(1+t+t^2) \cdots (1+t+\cdots+t^{n-1}) = \prod_{i=1}^n \frac{1-t^i}{1-t} = (1-t)^{-n} \prod_{i=1}^n (1-t^i)$$

From the additive isomorphism $H^*(F_n(\mathbb{C}^\infty); \mathbb{Z}) \approx H^*(G_n(\mathbb{C}^\infty); \mathbb{Z}) \otimes H^*(F_n(\mathbb{C}^n); \mathbb{Z})$ we see that the Poincaré series $p(t)$ of $H^*(G_n(\mathbb{C}^\infty); \mathbb{Z})$ satisfies

$$p(t)(1-t)^{-n} \prod_{i=1}^n (1-t^i) = (1-t)^{-n} \quad \text{and hence} \quad p(t) = \prod_{i=1}^n (1-t^i)^{-1}$$

This is exactly the Poincaré series of $\mathbb{Z}[\sigma_1, \dots, \sigma_n]$ since σ_i has degree i . As noted before, this implies that the image of p^* is all the symmetric polynomials.

This finishes the proof for $G_n(\mathbb{C}^\infty)$. The same arguments apply in the other two cases, using $\mathbb{Z}_2$ coefficients throughout in the real case and replacing ‘rank’ by ‘dimension’ for $\mathbb{Z}_2$ vector spaces. $\square$

These calculations show that the isomorphism $H^*(E; R) \approx H^*(B; R) \otimes_R H^*(F; R)$ of the Leray–Hirsch theorem is not generally a ring isomorphism, for if it were, then the polynomial ring $H^*(F_n(\mathbb{C}^\infty); \mathbb{Z})$ would contain a copy of $H^*(F_n(\mathbb{C}^n); \mathbb{Z})$ as a subring, but in the latter ring some power of every positive-dimensional element is zero since $H^k(F_n(\mathbb{C}^n); \mathbb{Z}) = 0$ for sufficiently large k .

The Gysin Sequence

Besides the Leray–Hirsch theorem, which deals with fiber bundles that are cohomologically like products, there is another special class of fiber bundles for which an

elementary analysis of their cohomology structure is possible. These are fiber bundles $S^{n-1} \rightarrow E \xrightarrow{p} B$ satisfying an orientability hypothesis that will always hold if B is simply-connected or if we take cohomology with $\mathbb{Z}_2$ coefficients. For such bundles we will show there is an exact sequence, called the **Gysin sequence**,

$$\cdots \rightarrow H^{i-n}(B; R) \xrightarrow{\smile e} H^i(B; R) \xrightarrow{p^*} H^i(E; R) \rightarrow H^{i-n+1}(B; R) \rightarrow \cdots$$

where e is a certain ‘Euler class’ in $H^n(B; R)$. Since $H^i(B; R) = 0$ for $i < 0$, the initial portion of the Gysin sequence gives isomorphisms $p^*: H^i(B; R) \xrightarrow{\cong} H^i(E; R)$ for $i < n-1$, and the more interesting part of the sequence begins

$$0 \rightarrow H^{n-1}(B; R) \xrightarrow{p^*} H^{n-1}(E; R) \rightarrow H^0(B; R) \xrightarrow{\smile e} H^n(B; R) \xrightarrow{p^*} H^n(E; R) \rightarrow \cdots$$

In the case of a product bundle $E = S^{n-1} \times B$ there is a section, a map $s: B \rightarrow E$ with $ps = \mathbb{1}$, so the Gysin sequence breaks up into split short exact sequences

$$0 \rightarrow H^i(B; R) \xrightarrow{p^*} H^i(S^{n-1} \times B; R) \rightarrow H^{i-n+1}(B; R) \rightarrow 0$$

which agrees with the Künneth formula $H^*(S^{n-1} \times B; R) \approx H^*(S^{n-1}; R) \otimes H^*(B; R)$. The splitting holds whenever the bundle has a section, even if it is not a product.

For example, consider the bundle $S^{n-1} \rightarrow V_2(\mathbb{R}^{n+1}) \xrightarrow{p} S^n$. Points of $V_2(\mathbb{R}^{n+1})$ are pairs (v_1, v_2) of orthogonal unit vectors in $\mathbb{R}^{n+1}$, and $p(v_1, v_2) = v_1$. If we think of v_1 as a point of S^n and v_2 as a unit vector tangent to S^n at v_1 , then $V_2(\mathbb{R}^{n+1})$ is exactly the bundle of unit tangent vectors to S^n . A section of this bundle is a field of unit tangent vectors to S^n , and such a vector field exists iff n is odd by Theorem 2.28. The fact that the Gysin sequence splits when there is a section then says that $V_2(\mathbb{R}^{n+1})$ has the same cohomology as the product $S^{n-1} \times S^n$ if n is odd, at least when $n > 1$ so that the base space S^n is simply-connected and the orientability hypothesis is satisfied. When n is even, the calculations at the end of §3.D show that $H^*(V_2(\mathbb{R}^{n+1}); \mathbb{Z})$ consists of $\mathbb{Z}$ ’s in dimensions 0 and $2n-1$ and a $\mathbb{Z}_2$ in dimension n . The latter group appears in the Gysin sequence as

$$\begin{array}{ccccccc} H^0(S^n) & \xrightarrow{\smile e} & H^n(S^n) & \longrightarrow & H^n(V_2(\mathbb{R}^{n+1})) & \longrightarrow & H^1(S^n) \\ \parallel & & \parallel & & \parallel & & \parallel \\ \mathbb{Z} & & \mathbb{Z} & & \mathbb{Z}_2 & & 0 \end{array}$$

hence the Euler class e must be twice a generator of $H^n(S^n)$ in the case that n is even. When n is odd it must be zero in order for the Gysin sequence to split.

This example illustrates a theorem in differential topology that explains why the Euler class has this name: The Euler class of the unit tangent bundle of a closed orientable smooth n -manifold M is equal to the Euler characteristic $\chi(M)$ times a generator of $H^n(M; \mathbb{Z})$.

Whenever a bundle $S^{n-1} \rightarrow E \xrightarrow{p} B$ has a section, the Euler class e must be zero from exactness of $H^0(B) \xrightarrow{\smile e} H^n(B) \xrightarrow{p^*} H^n(E)$ since p^* is injective if there is a section. Thus the Euler class can be viewed as an obstruction to the existence of a section: If the Euler class is nonzero, there can be no section. This qualitative